

KAAF UNIVERSITY COLLEGE



FACULTY OF HEALTH & ALLIED SCIENCES

DEPARTMENT OF NURSING

B.Sc. NURSING

REGULAR / WEEKEND

END OF FIRST SEMESTER EXAMINATIONS- 2023/2024

FIRST & SECOND YEAR (LEVEL 100 &200)

COURSE: KNUR 102: ANATOMY & PHYSIOLOGY I I/MARKING SCHEME

MAY, 2024

Duration: 2 HOURS

GENERAL INSTRUCTION TO STUDENTS:

1. Do not leave your seat unless you are given permission by an Invigilator
2. Please avoid the use of red and green ink.
3. Do not communicate in any way with any other candidate in the examination room.
4. All rough work must be written in the answer booklet provided. A line should be drawn through any rough work to indicate to the examiner that it is not part of the work to be marked. Writings should be done in pencils.
5. You should write legibly in black or blue ink
6. PDA's (Personal Digital Assistants) and mobile phones are NOT permitted in this examination.

7. You are not to leave the examination room until after the first hour of the examination.
8. You are not to leave the exam hall 30 minutes to the end of the examination

ANSWER THE FOLLOWING QUESTIONS IN THE VARIOUS SECTIONS

SECTION A

1. Which organ is responsible for filtering waste products from the blood and producing urine?

- A. Liver
- B. Kidney
- C. Bladder
- D. Urethra

Answer: B

2. Which structure carries urine from the kidney to the bladder?

- A. Ureter
- B. Urethra
- C. Nephron
- D. Renal pelvis

Answer: A

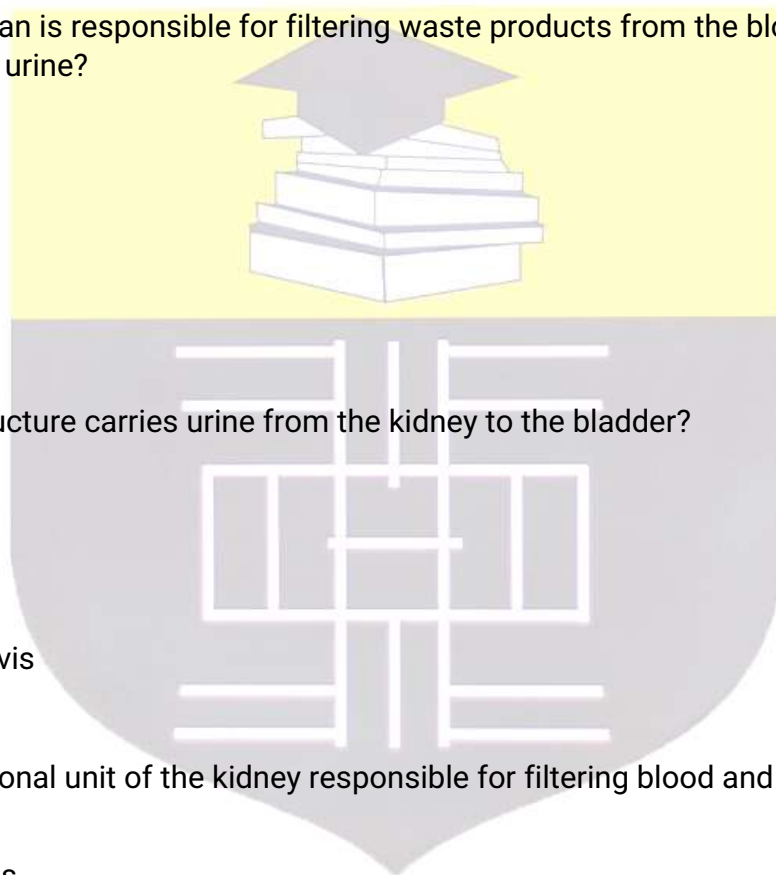
3. The functional unit of the kidney responsible for filtering blood and producing urine is called:

- A. Glomerulus
- B. Bowman's capsule
- C. Loop of Henle
- D. Nephron

Answer: D

4. What is the main function of the ureters?

- A. Storage of urine
- B. Filtration of blood



- C. Transport of urine from kidneys to bladder
- D. Regulation of blood pressure

Answer: C

5. Which part of the kidney collects urine and funnels it into the ureter?
- A. Renal medulla
 - B. Renal cortex
 - C. Renal pelvis
 - D. Renal pyramid

Answer: C

6. Which of the following is NOT a function of the urinary bladder?
- A. Storage of urine
 - B. Filtration of blood
 - C. Emptying urine into the urethra
 - D. Controlling the release of urine

Answer: B

7. What is the tube that carries urine from the bladder to the outside of the body?
- A. Ureter
 - B. Urethra
 - C. Nephron
 - D. Renal pelvis

Answer: B

8. The involuntary muscle responsible for controlling the release of urine from the bladder is called the:
- A. Sphincter
 - B. Renin
 - C. Glomerulus
 - D. Nephron

Answer: A

9. Which hormone regulates the reabsorption of water and electrolytes in the kidney tubules?

- A. Insulin
- B. Aldosterone
- C. Glucagon
- D. Thyroxine

Answer: B

10. Which of the following is NOT a symptom of kidney disease?

- A. Blood in urine
- B. Frequent urination
- C. Increased appetite
- D. Swelling of legs and ankles



Answer: C

11. What is the primary function of the male urethra?

- A. Transporting urine from the bladder to the outside of the body
- B. Producing sperm cells
- C. Carrying semen during ejaculation
- D. Filtering waste products from the blood

Answer: C

12. Which part of the male urethra passes through the prostate gland?

- A. Prostatic urethra
- B. Membranous urethra
- C. Spongy urethra
- D. Bulbous urethra

KNOWLEDGE AND EXCELLENCE

Answer: A

13. What is the terminal portion of the male urethra that extends through the penis?

- A. Prostatic urethra
- B. Membranous urethra
- C. Spongy urethra

D. Bulbous urethra

Answer: C

14. Where is the female bladder located in the body?

- A. Behind the liver
- B. Behind the stomach
- C. Behind the uterus and in front of the rectum
- D. Behind the kidneys

Answer: C

15. What is the primary function of the female bladder?

- A. Production of urine
- B. Storage of urine
- C. Filtration of blood
- D. Production of reproductive hormones

Answer: B

16. What is the tube that carries urine from the female bladder to the outside of the body?

- A. Ureter
- B. Urethra
- C. Nephron
- D. Renal pelvis

Answer: B

17. What is the micturition reflex?

- A. The reflex responsible for digestion
- B. The reflex responsible for regulating blood pressure
- C. The reflex responsible for the sensation of thirst
- D. The reflex responsible for urination

Answer: D

18. Which nervous system is primarily involved in the micturition reflex?

- A. Central nervous system
- B. Autonomic nervous system
- C. Peripheral nervous system
- D. Somatic nervous system

Answer: B

What triggers the micturition reflex?

- A. Stretching of the bladder walls
- B. Activation of the sympathetic nervous system
- C. Decrease in blood pressure
- D. Release of antidiuretic hormone (ADH)

Answer: A

19. Which of the following is NOT a component of the micturition reflex pathway?

- A. Spinal cord
- B. Bladder
- C. Brainstem
- D. Cerebral cortex

Answer: D

20. What happens during the micturition reflex?

- A. The bladder contracts and the external urethral sphincter relaxes
- B. The bladder relaxes and the external urethral sphincter contracts
- C. The bladder contracts and the internal urethral sphincter contracts
- D. The bladder relaxes and the internal urethral sphincter relaxes

Answer: A

21. Which of the following describes the location of the kidneys in the body?

- A. Retroperitoneal, lying on either side of the vertebral column
- B. Intraperitoneal, within the abdominal cavity
- C. Within the thoracic cavity, near the lungs

- D. Attached to the liver by ligaments

Answer: A

22. What is the functional unit of the kidney responsible for filtering blood and producing urine?

- A. Glomerulus
- B. Nephron
- C. Renal pelvis
- D. Ureter

Answer: B

23. Which region of the kidney contains the renal pyramids?

- A. Renal medulla
- B. Renal cortex
- C. Renal pelvis
- D. Renal capsule

Answer: A

24. What is the renal pelvis?

- A. A part of the nephron where filtration occurs
- B. The outer layer of the kidney
- C. A funnel-shaped structure that collects urine from the kidney tubules
- D. The portion of the kidney that contacts the adrenal glands

Answer: C

25. What is the structure that connects each kidney to the bladder and carries urine?

- A. Glomerulus
- B. Nephron
- C. Ureter
- D. Urethra

Answer: C

26. Which of the following organs primarily regulates fluid balance in the body?

- A. Liver

- B. Kidneys
- C. Pancreas
- D. Spleen

Answer: B) Kidneys

27. What is the primary electrolyte found outside the cells in the body?

- A. Sodium (Na^+)
- B. Potassium (K^+)
- C. Calcium (Ca^{2+})
- D. Magnesium (Mg^{2+})

Answer: A

28. Which hormone plays a key role in regulating sodium and water balance in the body by acting on the kidneys?

- A. Insulin
- B. Glucagon
- C. Aldosterone
- D. Thyroxine

Answer: C

29. Which condition is characterized by an excess of fluid in the body, often leading to swelling and edema?

- A. Dehydration
- B. Hypertension
- C. Hypernatremia
- D. Hypervolemia

Answer: D

30. What is the term for an electrolyte imbalance characterized by low levels of potassium in the blood?

- A. Hyperkalemia
- B. Hypokalemia
- C. Hyponatremia

D. Hypernatremia

Answer: B

Answer: True

31. True or False: Each testis is suspended within the scrotum by a structure called the epididymis.

Answer: False

32. Which of the following is the primary male reproductive organ responsible for producing sperm and testosterone?

- A. Prostate gland
- B. Testes
- C. Penis
- D. Epididymis

Answer: B

33. Where does spermatogenesis, the process of sperm production, primarily occur?

- A. Epididymis
- B. Seminal vesicles
- C. Vas deferens
- D. Seminiferous tubules

Answer: D

34. What structure carries sperm from the testes to the urethra during ejaculation?

- A. Vas deferens
- B. Epididymis
- C. Prostate gland
- D. Bulbourethral gland

Answer: A

35. What is the function of the prostate gland in the male reproductive system?

- A. Production of testosterone

- B. Storage of sperm
- C. Secretion of alkaline fluid to neutralize acidic vaginal pH
- D. Transport of sperm from the testes to the urethra

Answer: C

36. What is the purpose of the bulbourethral glands in the male reproductive system?

- A. Production of testosterone
- B. Neutralization of acidic urine in the urethra
- C. Storage of sperm
- D. Secretion of lubricating fluid during sexual arousal

Answer: D

READ THE QUESTIONS CAREFULLY AND FILL IN THE GAPS WITH THE CORRECT ANSWER

SECTION B

37. Case Scenario: A 35-year-old woman presents with irregular menstrual cycles and difficulty conceiving. On pelvic examination, she has enlarged ovaries with multiple fluid-filled cysts. This presentation is consistent with _____ syndrome.

Answer: polycystic ovary syndrome (PCOS)

38. Case Scenario: A 45-year-old woman complains of pelvic pain and heavy menstrual bleeding. Transvaginal ultrasound reveals the presence of an enlarged uterus with multiple leiomyomas. This condition is known as _____.

Answer: uterine fibroids

39. Case Scenario: A 28-year-old woman presents with lower abdominal pain and vaginal bleeding. On examination, she has an ectopic pregnancy confirmed by ultrasound. The most common site for ectopic pregnancy is the _____.

Answer: fallopian tube

40. Case Scenario: A 50-year-old woman complains of hot flashes, mood swings, and vaginal dryness. These symptoms are consistent with the onset of _____, a natural decline in reproductive hormones.

Answer: menopause

41. Case Scenario: A 30-year-old woman presents with severe dysmenorrhea and pelvic pain. On pelvic examination, tender nodules are palpated along the posterior vaginal

wall. This is suggestive of _____.

Answer: endometriosis

42. The primary functional unit of the kidney responsible for urine formation is the _____.

Answer: nephron

43. Filtration of blood occurs in the renal corpuscle, which consists of the glomerulus and the _____.

Answer: Bowman's capsule

44. Urine analysis involves the examination of _____ to assess various aspects of kidney function and overall health.

Answer: urine

45. One of the most common tests in urine analysis is the evaluation of urine _____, which can indicate hydration status and kidney function.

Answer: color

46. A urine sample with a high specific gravity may indicate _____.

Answer: dehydration

47. The presence of protein in urine, known as _____, may indicate kidney damage or disease.

Answer: proteinuria

48. The presence of glucose in urine, known as _____, may indicate diabetes mellitus.

Answer: glucosuria

49. A urine test that detects the presence of bacteria or white blood cells may indicate a urinary tract _____.

Answer: infection

50. Case Scenario: A 30-year-old woman presents with excessive thirst and frequent urination. Laboratory tests reveal low levels of antidiuretic hormone (ADH). This condition, known as _____, results from a deficiency of ADH secretion from the posterior pituitary gland.

Answer: diabetes insipidus

51. Case Scenario: A 40-year-old man experiences severe headaches, visual

disturbances, and fatigue. MRI imaging reveals a tumor compressing the pituitary gland. The tumor likely affects the secretion of hormones, including _____, which is synthesized in the hypothalamus and released by the posterior pituitary gland.

Answer: oxytocin

52. Case Scenario: A 25-year-old woman presents with postpartum hemorrhage. Administration of an oxytocin analog helps stimulate uterine contractions, aiding in the prevention of excessive bleeding. This medication acts on receptors in the uterus, mimicking the effects of natural _____ released by the posterior pituitary gland.

Answer: oxytocin

53. Case Scenario: A 55-year-old man complains of fatigue, weight gain, and cold intolerance. Laboratory tests reveal low levels of thyroid hormones, along with low levels of TSH (thyroid-stimulating hormone). Further investigation reveals a tumor affecting the hypothalamus-pituitary-thyroid axis, leading to decreased secretion of _____ from the posterior pituitary gland.

Answer: thyroid-stimulating hormone (TSH)

54. Case Scenario: A 35-year-old woman presents with severe uterine cramps during menstruation. She reports a family history of menstrual irregularities. Further evaluation reveals normal levels of estrogen and progesterone but decreased levels of prostaglandins. Dysfunction of the hypothalamus-pituitary-ovary axis or abnormalities in the release of _____ from the posterior pituitary gland may contribute to her symptoms.

Answer: oxytocin

55. Growth hormone deficiency in children can lead to _____, characterized by short stature and delayed growth.

Answer: dwarfism

56. The _____ is a fatty, insulating layer that surrounds the axon of many neurons, enhancing the speed of electrical signal transmission.

Answer: myelin sheath

57. The myelin sheath is primarily formed by specialized glial cells called _____ in the central nervous system and _____ in the peripheral nervous system.

Answer: oligodendrocytes; Schwann cells

58. Damage to the myelin sheath can result in neurological conditions such as _____, characterized by impaired motor function and sensory deficits.

Answer: multiple sclerosis

59. Action potentials are generated when the membrane potential of a neuron reaches a threshold level, typically around _____ millivolts.
60. During the depolarization phase of an action potential, voltage-gated _____ channels open, allowing an influx of sodium ions into the neuron, causing further depolarization.

Answer: sodium

61. Following the depolarization phase, voltage-gated _____ channels open, allowing an efflux of potassium ions out of the neuron, leading to repolarization of the membrane potential.

Answer: potassium

62. In the process of simple propagation of nerve impulses, once the action potential reaches the axon terminal, it triggers the opening of voltage-gated _____ channels.

Answer: calcium

63. Case Scenario: A person accidentally touches a hot stove, resulting in immediate withdrawal of the hand. This reflex action, known as the _____ reflex, is mediated by the spinal cord without involvement of the brain.

Answer: withdrawal

64. Case Scenario: During a routine medical check-up, the doctor taps the patellar tendon with a reflex hammer, causing the leg to kick forward. This reflex, known as the _____ reflex, tests the integrity of the spinal cord and nerve pathways involved in knee-jerk responses.

Answer: patellar

65. Case Scenario: A newborn infant exhibits a sucking reflex when the lips are touched. This reflex, essential for feeding, is initiated by stimulation of sensory receptors in the lips and transmitted through the _____ nerves to the spinal cord, resulting in the rhythmic sucking motion.

Answer: cranial

66. Which of the following conditions is characterized by the abnormal growth of cells in the prostate gland, leading to urinary symptoms such as difficulty urinating?
- A. Erectile dysfunction
 - B. Prostate cancer
 - C. Benign prostatic hyperplasia (BPH)

D. Testicular torsion

Answer: C

67. Which of the following conditions is a common cause of infertility in women and is characterized by the inability to ovulate regularly?

- A. Endometriosis
- B. Polycystic ovary syndrome (PCOS)
- C. Fibroids
- D. Pelvic inflammatory disease (PID)

Answer: B

68. Which of the following conditions is a medical emergency in males characterized by the twisting of the spermatic cord, leading to reduced blood flow to the testicle?

- A. Erectile dysfunction
- B. Testicular cancer
- C. Epididymitis
- D. Testicular torsion

Answer: D

69. Which of the following hormones is NOT produced by the anterior pituitary gland?

- A. Growth hormone (GH)
- B. Prolactin (PRL)
- C. Thyroid-stimulating hormone (TSH)
- D. Antidiuretic hormone (ADH)

Answer: D

70. What is the primary function of adrenocorticotrophic hormone (ACTH) produced by the anterior pituitary gland?

- A. Regulation of growth and development
- B. Stimulation of milk production
- C. Regulation of thyroid function
- D. Stimulation of cortisol production by the adrenal glands

Answer: D

71. Which hormone stimulates milk production in the mammary glands?

- A. Follicle-stimulating hormone (FSH)
- B. Luteinizing hormone (LH)
- C. Prolactin (PRL)
- D. Thyroid-stimulating hormone (TSH)

Answer: C

72. What is the function of thyroid-stimulating hormone (TSH) produced by the anterior pituitary gland?

- A. Regulation of thyroid hormone production and release
- B. Stimulation of growth hormone production
- C. Stimulation of cortisol production
- D. Regulation of blood calcium levels

Answer: A

73. Which hormone stimulates ovulation in females and testosterone production in males?

- A. Growth hormone (GH)
- B. Luteinizing hormone (LH)
- C. Prolactin (PRL)
- D. Adrenocorticotrophic hormone (ACTH)

Answer: B

74. What is the primary function of growth hormone (GH) produced by the anterior pituitary gland?

- A. Regulation of metabolism
- B. Stimulation of bone growth and muscle development
- C. Stimulation of milk production
- D. Regulation of blood pressure

Answer: B

75. Which hormone stimulates the growth and development of ovarian follicles in females and sperm production in males?

- A. Follicle-stimulating hormone (FSH)
- B. Luteinizing hormone (LH)
- C. Prolactin (PRL)
- D. Adrenocorticotrophic hormone (ACTH)

Answer: A

76. Which hormone is involved in the regulation of water balance in the body by promoting water reabsorption in the kidneys?

- A. Growth hormone (GH)
- B. Prolactin (PRL)
- C. Thyroid-stimulating hormone (TSH)
- D. Antidiuretic hormone (ADH)

Answer: D

77. What is the function of prolactin (PRL) produced by the anterior pituitary gland?

- A. Stimulation of milk production
- B. Regulation of thyroid function
- C. Stimulation of cortisol production
- D. Stimulation of growth hormone production

Answer: A

78. Which hormone stimulates the growth and development of ovarian follicles in females and spermatogenesis in males?

- A. Follicle-stimulating hormone (FSH)
- B. Luteinizing hormone (LH)
- C. Prolactin (PRL)
- D. Thyroid-stimulating hormone (TSH)

Answer: A

79. Which of the following hormones is primarily responsible for regulating metabolism and energy expenditure in the body?

- A. Insulin
- B. Growth hormone (GH)

- C. Thyroid-stimulating hormone (TSH)
- D. Triiodothyronine (T3) and Thyroxine (T4)

Answer: D

80. What is the primary function of T3 (triiodothyronine) and T4 (thyroxine) hormones?

- A. Regulation of blood glucose levels
- B. Stimulation of bone growth
- C. Regulation of metabolic rate and body temperature
- D. Control of calcium homeostasis

Answer: C

81. Which gland secretes T3 (triiodothyronine) and T4 (thyroxine) hormones?

- A. Adrenal gland
- B. Pituitary gland
- C. Thyroid gland
- D. Pancreas

Answer: C

82. Which of the following hormones stimulates the production and release of T3 and T4 from the thyroid gland?

- A. Insulin
- B. Growth hormone (GH)
- C. Thyroid-stimulating hormone (TSH)
- D. Parathyroid hormone (PTH)

Answer: C

83. What is the primary difference between T3 and T4 hormones?

- A. T3 has three iodine atoms, while T4 has four iodine atoms.
- B. T3 is produced by the adrenal glands, while T4 is produced by the thyroid gland.
- C. T3 is more potent and has a faster metabolic effect than T4.
- D. T4 is released in larger quantities compared to T3.

Answer: A

84. The central nervous system (CNS) consists of which two main components?

- A. Brain and spinal cord
- B. Brain and peripheral nerves
- C. Spinal cord and peripheral nerves
- D. Brainstem and cerebellum

Answer: A

85. Which part of the nervous system is responsible for transmitting sensory information from the body to the brain and sending motor commands from the brain to the muscles and glands?

- A. Central nervous system (CNS)
- B. Peripheral nervous system (PNS)
- C. Autonomic nervous system (ANS)
- D. Enteric nervous system (ENS)

Answer: B

86. The autonomic nervous system (ANS) is further divided into which two branches?

- A. Sympathetic and somatic
- B. Parasympathetic and somatic
- C. Sympathetic and parasympathetic
- D. Central and peripheral

Answer: C

87. Which part of the nervous system is responsible for involuntary functions such as heart rate, digestion, and breathing?

- A. Central nervous system (CNS)
- B. Peripheral nervous system (PNS)
- C. Autonomic nervous system (ANS)
- D. Somatic nervous system (SNS)

Answer: C

88. The somatic nervous system (SNS) is responsible for controlling which type of activities?

- A. Voluntary activities such as walking and talking
- B. Involuntary activities such as heart rate and digestion
- C. Emotional responses such as fear and anxiety
- D. Internal organ functions such as breathing and blood pressure

Answer: A

89. Which division of the peripheral nervous system is responsible for carrying motor signals from the central nervous system to muscles and glands?

- A. Somatic nervous system
- B. Autonomic nervous system
- C. Sympathetic nervous system
- D. Parasympathetic nervous system

Answer: A

90. Which neurotransmitter is commonly released at the neuromuscular junction in the somatic motor system to stimulate muscle contraction?

- A. Acetylcholine
- B. Dopamine
- C. Serotonin
- D. Norepinephrine

Answer: A

91. The somatic motor system primarily controls which type of muscle?

- A. Smooth muscle
- B. Cardiac muscle
- C. Skeletal muscle
- D. Visceral muscle

Answer: C

92. Which of the following is NOT a function of the somatic nervous system?

- A. Voluntary control of skeletal muscles

- B. Regulation of internal organ functions
- C. Sensory perception of touch, pain, and temperature
- D. Reflex responses to stimuli

Answer: B

93. Which neurotransmitter is commonly released at the neuromuscular junction in the somatic nervous system to stimulate skeletal muscle contraction?

- A. Acetylcholine
- B. Dopamine
- C. Serotonin
- D. Norepinephrine

Answer: A

94. Damage to the somatic nervous system can lead to impairment in which of the following?

- A. Vision and hearing
- B. Digestion and metabolism
- C. Voluntary movement and coordination
- D. Heart rate and blood pressure

Answer: C

95. Which of the following is the primary function of neurons?

- A. Regulation of hormone secretion
- B. Conduction of electrical impulses
- C. Storage of energy
- D. Maintenance of bone density

Answer: B

96. What is the role of dendrites in a neuron?

- A. Transmit electrical impulses away from the cell body
- B. Receive signals from other neurons or sensory receptors
- C. Produce and release neurotransmitters
- D. Provide structural support to the neuron

Answer: B

97. Which part of the neuron contains the nucleus and most of the organelles?

- A. Axon
- B. Dendrite
- C. Cell body (soma)
- D. Myelin sheath

Answer: C

98. What is the function of the myelin sheath?

- A. To insulate the axon and speed up the transmission of electrical impulses
- B. To anchor the neuron in place
- C. To release neurotransmitters into the synaptic cleft
- D. To receive signals from other neurons

Answer: A

99. Which type of neuron carries signals from sensory receptors to the central nervous system (CNS)?

- A. Motor neuron
- B. Interneuron
- C. Sensory neuron
- D. Afferent neuron

Answer: C

100. What is saltatory conduction?

- A. Slow conduction of action potentials along unmyelinated axons
- B. Fast conduction of action potentials along myelinated axons, jumping from one node of Ranvier to the next
- C. The transmission of electrical signals across a synapse
- D. The process of depolarization and repolarization of the neuron membrane

Answer: B

101. What is the refractory period of a neuron?
- A. The period during which a neuron is more sensitive to stimulation
 - B. The period during which a neuron cannot generate another action potential
 - C. The period during which a neuron is at rest and not conducting any electrical activity
 - D. The period during which a neuron releases neurotransmitters into the synaptic cleft

Answer: B

102. Which of the following is true regarding the refractory period?
- A. It ensures that action potentials propagate backward along the axon
 - B. It prevents action potentials from being generated too frequently and allows for discrete signaling
 - C. It speeds up the conduction of action potentials along the axon
 - D. It occurs only in myelinated axons

Answer: B

103. Which of the following is NOT a type of neuroglia cell in the central nervous system (CNS)?
- A. Astrocyte
 - B. Oligodendrocyte
 - C. Microglia
 - D. Schwann cell

Answer: D

104. Which neuroglia cell type is responsible for providing structural support, regulating ion and neurotransmitter levels, and forming the blood-brain barrier in the CNS?
- A. Oligodendrocyte
 - B. Astrocyte
 - C. Microglia
 - D. Ependymal cell

Answer: B

105. What is the primary function of oligodendrocytes in the CNS?

- A. Phagocytosis of debris and pathogens
- B. Formation of the blood-brain barrier
- C. Production of myelin sheath around axons
- D. Regulation of cerebrospinal fluid (CSF) production

Answer: C

106. Which neuroglia cell type is considered the resident immune cells of the CNS, responsible for detecting and removing pathogens and debris?

- A. Astrocyte
- B. Oligodendrocyte
- C. Microglia
- D. Ependymal cell



Answer: C

107. Ependymal cells in the CNS are primarily involved in which of the following functions?

- A. Providing structural support to neurons
- B. Forming myelin sheath around axons
- C. Producing cerebrospinal fluid (CSF)
- D. Regulating ion and neurotransmitter levels

Answer: C

108. What is the primary function of the spinal cord?

- A. Processing sensory information
- B. Generating motor commands
- C. Relay station for olfactory signals
- D. Regulation of hormone secretion

Answer: A

109. How many pairs of spinal nerves are connected to the spinal cord?

- A. 8
- B. 12
- C. 31

D. 42

Answer: C

110. Which part of the nervous system is responsible for transmitting sensory information to the spinal cord?

- A. Afferent nervous system
- B. Efferent nervous system
- C. Central nervous system
- D. Peripheral nervous system

Answer: D

111. How many pairs of spinal nerves are there in the human body?

- A. 12
- B. 24
- C. 31
- D. 40

Answer: C

112. Spinal nerves emerge from the spinal cord through which structure?

- A. Dorsal root ganglion
- B. Ventral root ganglion
- C. Intervertebral foramen
- D. Epidural space

Answer: C

113. Spinal nerves are classified as:

- A. Sensory only
- B. Motor only
- C. Mixed (containing both sensory and motor fibers)
- D. Autonomic

Answer: C

114. The dorsal root of a spinal nerve primarily carries which type of information?

- A. Motor commands

- B. Sensory information
- C. Autonomic signals
- D. None of the above

Answer: B

115. The ventral root of a spinal nerve primarily carries which type of information?

- A. Motor commands
- B. Sensory information
- C. Autonomic signals
- D. None of the above

Answer: A

116. Which structure forms when the dorsal and ventral roots of a spinal nerve converge?

- A. Dorsal ramus
- B. Ventral ramus
- C. Dorsal root ganglion
- D. Spinal nerve plexus

Answer: D

117. Spinal nerves that emerge from the cervical region form part of which plexus?

- A. Cervical plexus
- B. Brachial plexus
- C. Lumbar plexus
- D. Sacral plexus

Answer: A

118. The sciatic nerve, one of the largest nerves in the body, originates from which plexus?

- A. Cervical plexus
- B. Brachial plexus
- C. Lumbar plexus
- D. Sacral plexus

Answer: D

119. Damage to which nerve plexus could result in weakness or paralysis of the upper limb
- A. Cervical plexus
 - B. Brachial plexus
 - C. Lumbar plexus
 - D. Sacral plexus

Answer: B

120. Spinal nerves arising from the lumbar region contribute to which plexus?
- A. Cervical plexus
 - B. Brachial plexus
 - C. Lumbar plexus
 - D. Sacral plexus

Answer: C) Lumbar plexus

121. Which of the following is NOT a common cause of spinal cord injury?
- A. Motor vehicle accidents
 - B. Falls
 - C. Sports injuries
 - D. Bacterial infections

Answer: D

122. Damage to the spinal cord can result in loss of:
- A. Motor function only
 - B. Sensory function only
 - C. Both motor and sensory function
 - D. Cognitive function

Answer: C

123. Which part of the brain is responsible for regulating basic functions such as breathing, heart rate, and sleep-wake cycles?
- A. Cerebrum

- B. Cerebellum
- C. Brainstem
- D. Limbic system

Answer: C

124. The primary motor cortex, responsible for voluntary muscle movement, is located in which lobe of the brain?

- A. Frontal lobe
- B. Parietal lobe
- C. Occipital lobe
- D. Temporal lobe

Answer: A

125. Which part of the brain is associated with the processing of sensory information such as touch, temperature, and pain?

- A. Parietal lobe
- B. Frontal lobe
- C. Occipital lobe
- D. Temporal lobe

Answer: A

126. The hippocampus, a part of the limbic system, plays a crucial role in which brain function?

- A. Memory formation and retrieval
- B. Motor coordination
- C. Visual processing
- D. Language comprehension

Answer: A

127. Which part of the brain is responsible for coordinating movement, maintaining posture, and balance?

- A. Cerebrum
- B. Cerebellum
- C. Brainstem

D. Thalamus

Answer: B

128. The visual cortex, responsible for processing visual information, is located in which lobe of the brain?

- A. Frontal lobe
- B. Parietal lobe
- C. Occipital lobe
- D. Temporal lobe

Answer: C

129. Broca's area, involved in speech production, is typically located in which hemisphere of the brain?

- A. Right hemisphere
- B. Left hemisphere
- C. Frontal lobe
- D. Temporal lobe

Answer: B

130. Which part of the brain is responsible for relaying sensory and motor signals to and from the cerebral cortex?

- A. Thalamus
- B. Hypothalamus
- C. Amygdala
- D. Basal ganglia

Answer: A

131. The prefrontal cortex, located in the frontal lobe, is associated with which cognitive functions?

- A. Decision-making, planning, and social behavior
- B. Visual processing and spatial awareness
- C. Language comprehension and production
- D. Memory formation and retrieval

Answer: A

132. Damage to which part of the brain is associated with the loss of language comprehension and production, known as aphasia? A) Occipital lobe B) Temporal lobe C) Frontal lobe D) Wernicke's area

Answer: D

133. Which of the following statements best describes the function of the thalamus?
- A. The thalamus regulates body temperature and hunger.
 - B. The thalamus serves as a relay station for sensory information.
 - C. The thalamus controls voluntary muscle movements.
 - D. The thalamus produces hormones involved in stress response.

Answer: B

134. Damage to the thalamus is most likely to result in:
- A. Loss of memory and cognitive function.
 - B. Impaired coordination and balance.
 - C. Altered perception of sensory stimuli.
 - D. Difficulty in regulating emotions and mood.

Answer: C

135. Which of the following is NOT a function of the thalamus?
- A. Regulation of sleep-wake cycles
 - B. Integration of sensory information
 - C. Relay of motor commands to the spinal cord
 - D. Control of autonomic functions such as heartbeat and breathing

Answer: D

136. Which hormone is primarily produced and released by the hypothalamus to regulate the secretion of hormones from the anterior pituitary gland?
- A. Thyroid-stimulating hormone (TSH)
 - B. Growth hormone (GH)
 - C. Gonadotropin-releasing hormone (GnRH)

D. Oxytocin

Answer: C

137. The hypothalamus plays a crucial role in regulating which of the following physiological processes?

- A. Blood clotting
- B. Calcium homeostasis
- C. Sleep-wake cycles
- D. Oxygen transport in the blood

Answer: C

138. Which sense organ is primarily responsible for detecting sound waves and converting them into electrical signals for the brain to interpret?

- A. Eyes
- B. Ears
- C. Nose
- D. Tongue

Answer: B

139. Which sense organ is responsible for detecting chemical molecules in the air and translating them into the perception of smell?

- A. Eyes
- B. Ears
- C. Nose
- D. Skin

Answer: C

140. The sense organ of taste primarily detects which of the following?

- A. Light waves
- B. Sound waves
- C. Chemical molecules dissolved in saliva
- D. Mechanical pressure

Answer: C

141. Which sense organ is responsible for detecting light waves and translating them into visual signals for the brain?

- A. Ears
- B. Nose
- C. Eyes
- D. Tongue

Answer: C

142. The sense organ responsible for detecting touch, pressure, temperature, and pain is primarily located in which part of the body?

- A. Skin
- B. Eyes
- C. Ears
- D. Nose

Answer: A

143. The primary function of the lymphatic system is to _____ excess tissue fluid and return it to the bloodstream.

Answer: drain

Answer: lymphatic ducts

LABEL THE DIAGRAM OF THE HUMAN BRAIN

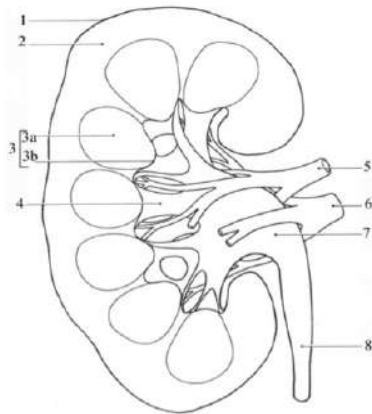
SECTION C

144. What is number 1?



LECTURER: MRS AMAKU CALYS-TAGOE
NUMBER:.....

STUDENT INDEX



145. Number 2 indicates.....

146. Number 3 indicates a)..... b).....

147. What is number 5?.....

148. What is number 6?.....

149. What is number 7?.....

150. What is number 8?.....

GOOD LUCK!!!!

KNOWLEDGE AND EXCELLENCE